

The T-Shaped Matching Network

This network is called T because its configuration resembles the Latin letter T. In this configuration, two L-shaped networks with a shared leg are placed adjacently. At the intersection of the two L-shaped networks, a single virtual resistance is considered for this design. The virtual resistor is **larger** than R_L and R_S .

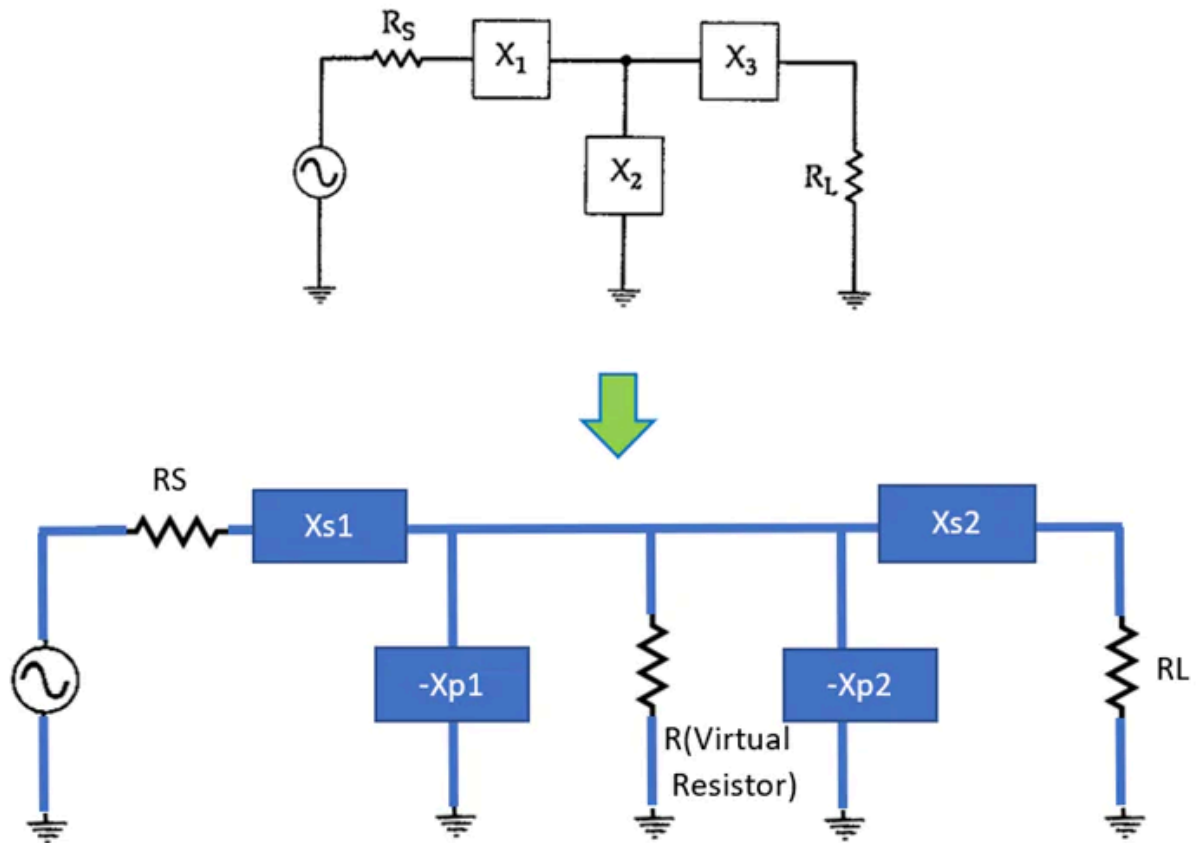
If termination impedances are high, $(R_S, R_L) > 100 \Omega$, the Pi network should be implemented. T-shaped networks are applicable if the termination impedances are low, $(R_S, R_L) < 100 \Omega$.

$$Q = \sqrt{\frac{R}{R_{smaller}} - 1}$$

$R_{smaller}$ = the smallest termination resistor

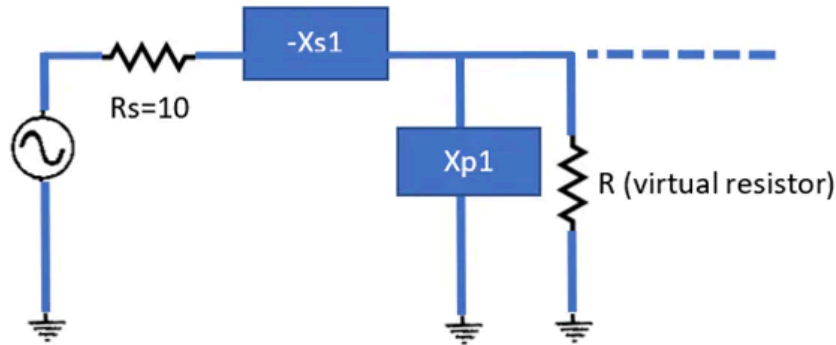
R = virtual resistor

The virtual resistor can be defined based on your desired Q of the circuit.



Then the movement between load and source will be divided into two sections ;

1) From source to virtual resistor (as the source is smaller than load)

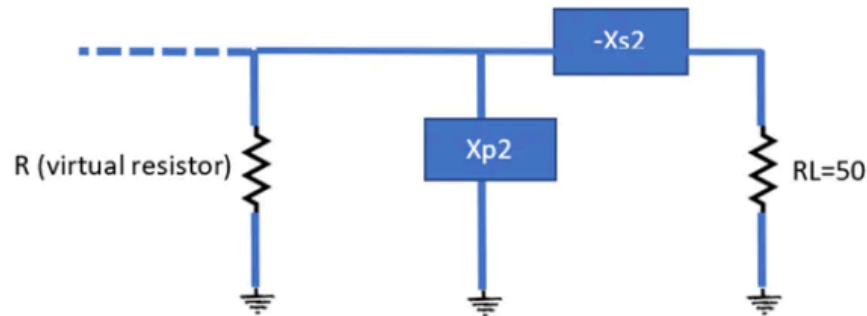


$$R = R_{small}(Q^2 + 1) \quad R = 10(10^2 + 1) = 1010 \, \Omega$$

$$X_{p1} = \frac{R}{Q} = \frac{1010}{10} = 101 \, \Omega$$

$$X_{s1} = Q \times R_s = 10 \times 10 = 100 \, \Omega$$

2) From virtual resistor to the load

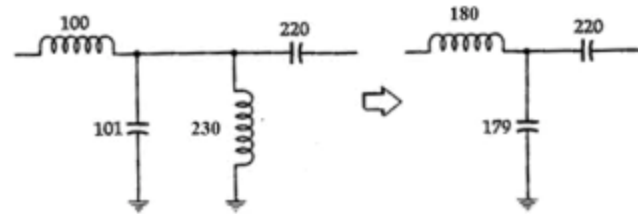


$$Q2 = \sqrt{\frac{R}{R_L} - 1} = \sqrt{\frac{1010}{50} - 1} = 4.4$$

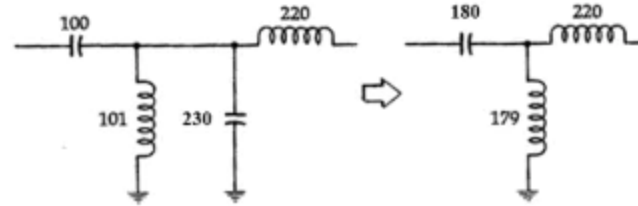
$$X_{p2} = \frac{R}{Q2} = \frac{1010}{4.4} = 230 \, \Omega$$

$$X_{s2} = Q2 \times R_L = 4.4 \times 50 = 220 \, \Omega$$

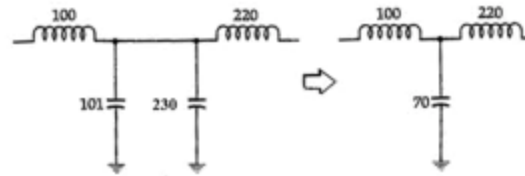
The arrangement of the designed T network can be like each of these below structures:



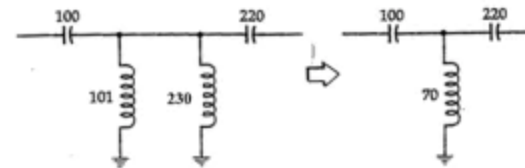
(C)



(D)



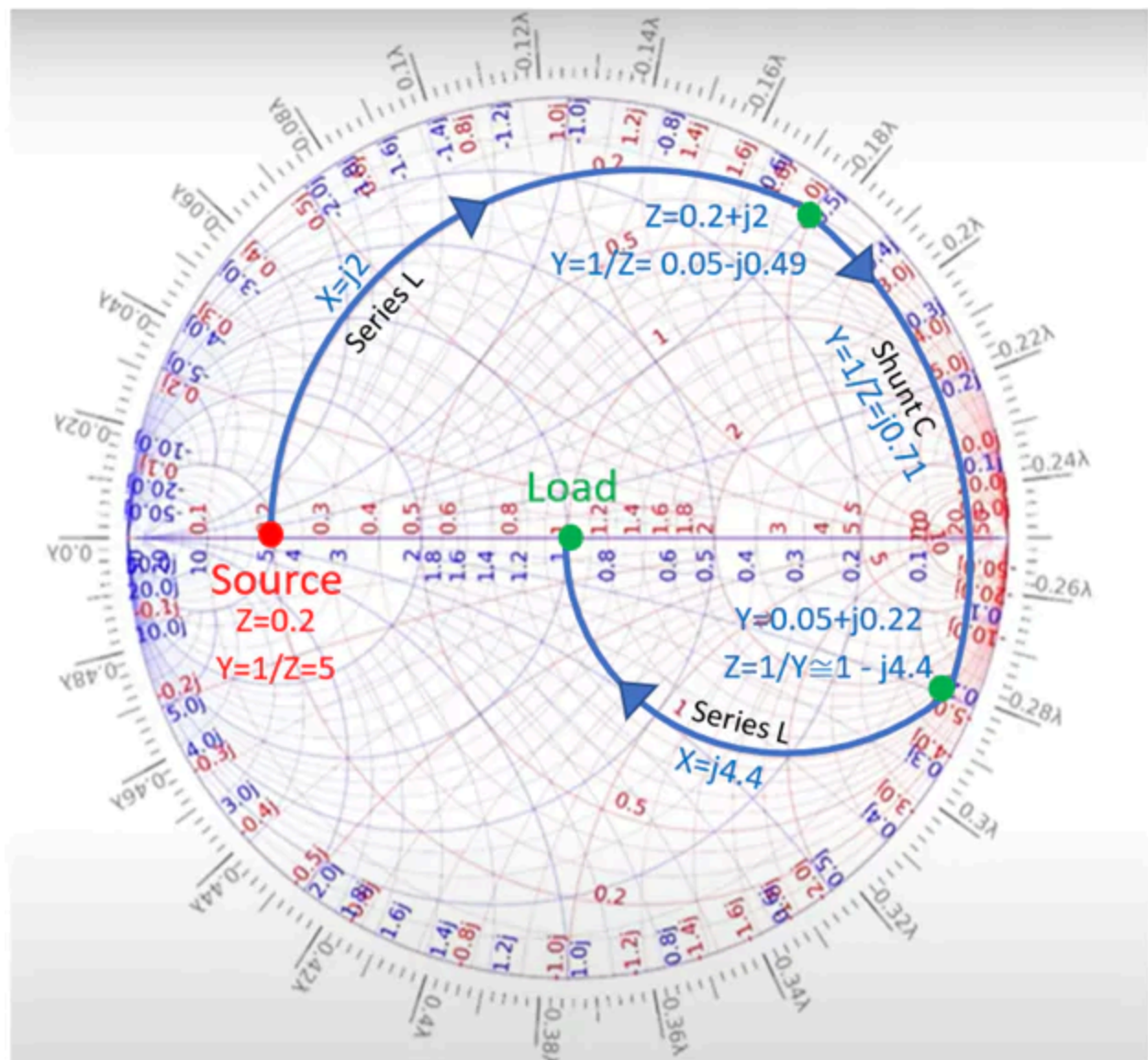
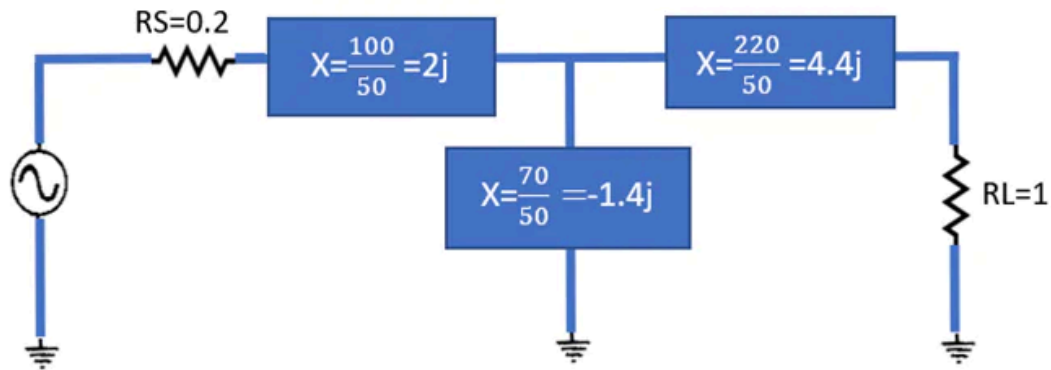
(A)



(B)

Applying the values to the smith chart for the first T matching network:

- First normalize the values by dividing them to the characteristic impedance; $Z_0 = 50 \Omega$
- Determine the place of load and source impedances on the Smith chart
- Move from source to the load
- For the shunt elements their **admittance** must be added to the previous admittance and for the series elements the **impedance** must be added to the previous impedance.



$$j100 = j\omega L = j(2\pi f)L; \text{ if } f = 13.56 \text{ MHz}; L = 1.17 \mu\text{H}$$

$$j220 = j\omega L = j(2\pi f)L; \text{ if } f = 13.56 \text{ MHz}; L = 2.58 \mu\text{H}$$

$$-j70 = \frac{1}{j\omega C} = \frac{1}{j(2\pi f)C}; \text{ if } f = 13.56 \text{ MHz}; C = 167 \text{ PF}$$

The T and Pi networks are ideal for **narrow-band** and **high-Q**. As we discussed before, the Q for the L-shaped matching network has the low value, allowing for wide bandwidth to be achieved [1]. We can also increase the bandwidth by employing an L-shaped series.

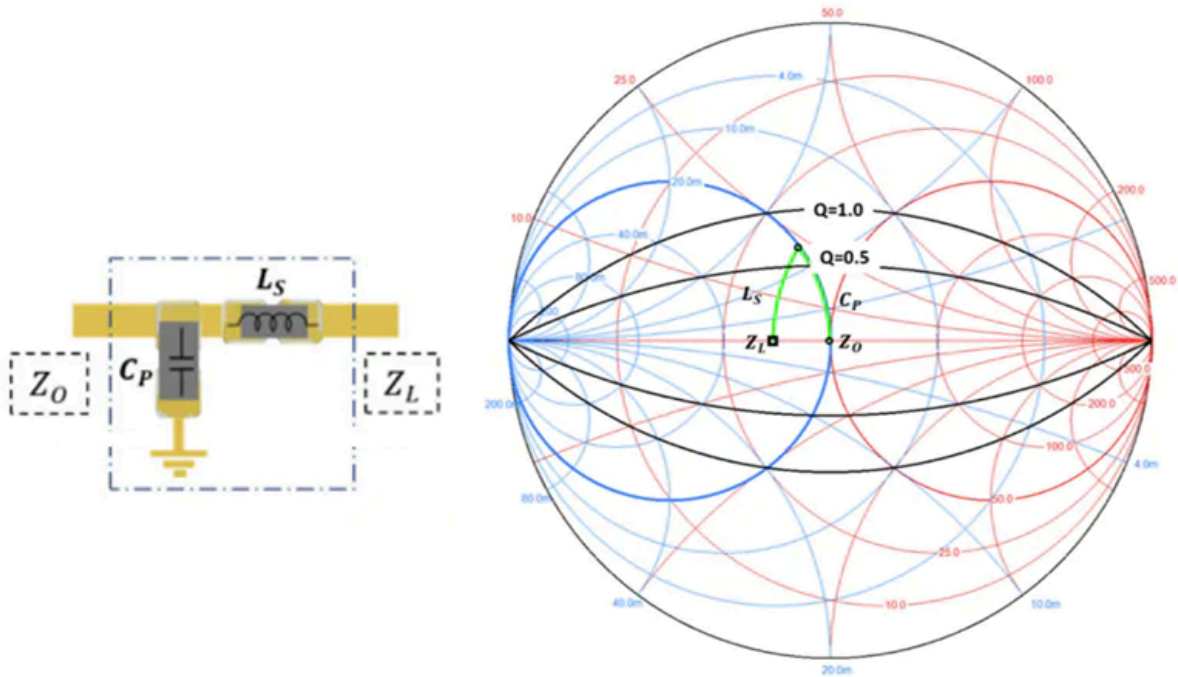
Comparison between an L-shaped matching network and an L-shaped series

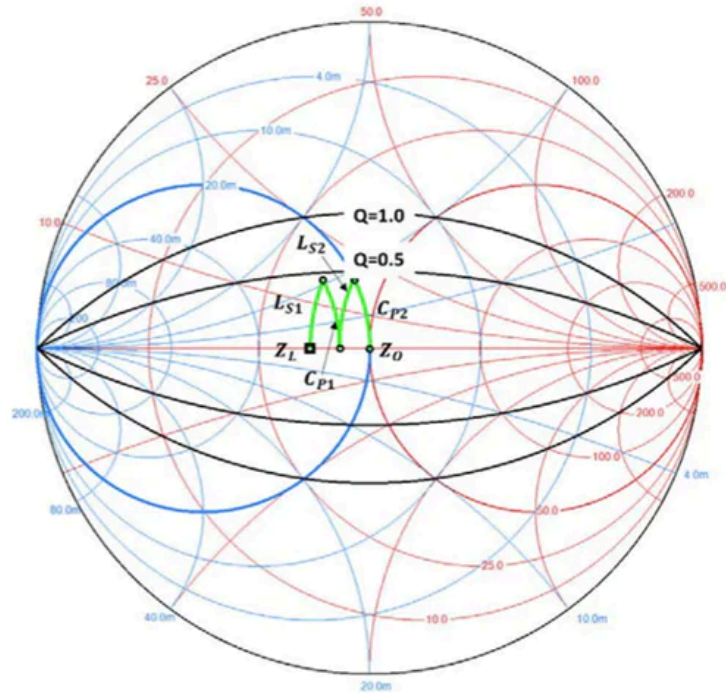
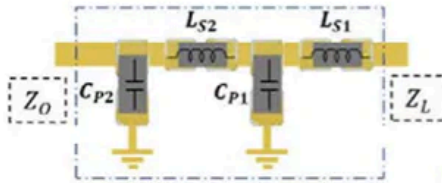
The movement from Z_{load} to Z_{source} is accomplished with a single L-shaped matching network, as shown below. As the movement from load to source is on impedance circles and upwards, the series inductor is added first. The subsequent movement is on inductance circles and downwards, so the parallel capacitor is added, the Q is 0.7 for this network.

The four-element L-shaped network can be used for the previous example to enhance the bandwidth. Here, the Q is smaller (0.5), allowing for a wider bandwidth. As a result, more series L-shaped matching networks could be considered to increase the bandwidth [2]. The optimum value for Q is achieved by:

$$Q = \sqrt{\frac{R}{R_{\text{smaller}}}} - 1 = \sqrt{\frac{R_{\text{larger}}}{R}} - 1$$

Where R is the virtual resistance ; and R_{smaller} are R_{larger} the smallest and largest resistance values. Notably, the Q of each network that operates independently will be higher than when the repetition of networks is considered.





Conclusion

The simplest matching network is the two-element L-shaped one with lowest Q (quality factor) and wide bandwidth. The Q in an L-shaped matching network could not be altered because it is defined by the specific values of the load and source impedances. However, the Pi and T matching networks can be utilized to achieve the desired Q . The Pi network is utilized for high impedances ($R_S, R_L > 100\Omega$) while the T network is utilized for low impedances ($R_S, R_L < 100\Omega$). These two networks have high Q , and are ideal for narrow band matching networks. Notably, a series of L-shaped matching networks can be employed to provide a circuit with wider bandwidth.

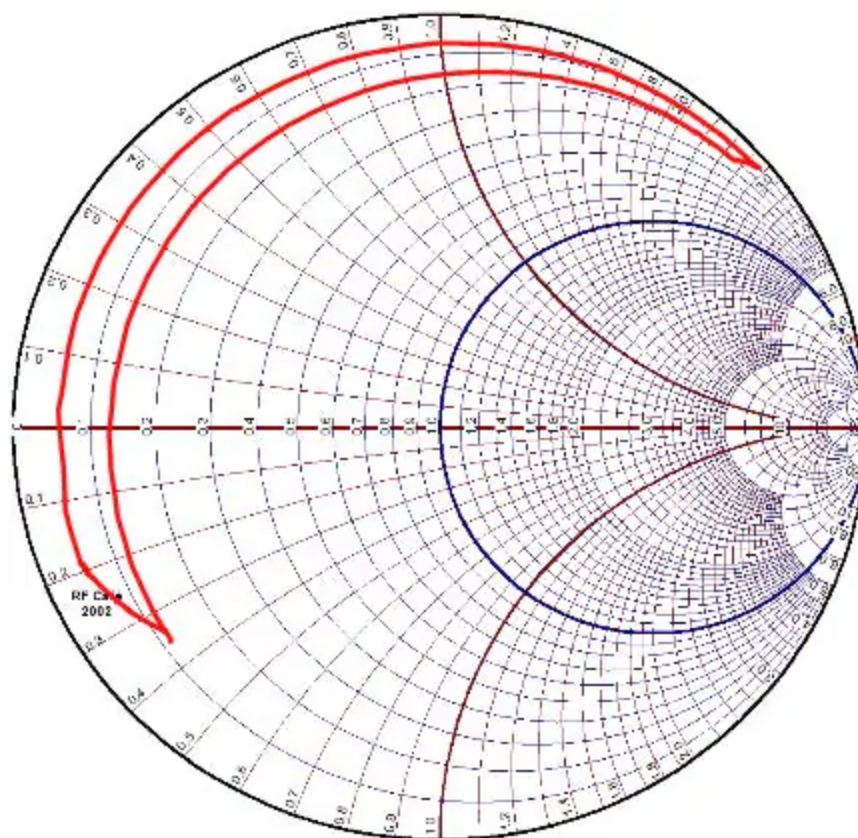
Characterize Your Matching Unit

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